

Automatic Factor Grouping

Exposure deduplication and latent return factors

Abstract

This cluster-only run groups 100 admitted signals without known-factor labels. A training-only economic-admission stage retained 100 of 140 coverage-valid signals before HDBSCAN. HDBSCAN first collapses signals that rank the same stocks, including sign-reversed definitions. The resulting primitive FMP returns are then decomposed by parallel-analysis PCA and varimax rotation. The fresh version=20260829T051952Z sample produces 5 final candidate factors. Each parent gives every signed primitive composite one equal vote, regardless of how many raw signals that primitive contains. No post-clustering signal elimination or factor-admission rule is applied.

1 Sample and method

Setting	Value
Gold	version=20260829T051952Z
Requested symbols	all
Panel symbols	3014
Exposure window	2012-01-01 to 2022-12-31
Raw price floor	\$1.00
Market-cap screen	top 90.0% of names
Financials excluded	true
Minimum names / usable month	20
Minimum usable months	24
Pre-cluster Sharpe admission	true
Admission probability threshold	70.0%
Admission bootstrap repetitions	2000
Admission expected block months	12
HDBSCAN minimum cluster size	2
Block-bootstrap repetitions	100
Primitive FMP sector demeaning	sector
Primitive FMP beta neutrality	false
Final factor FMP sector demeaning	sector
Final factor FMP beta neutrality	true
Parallel-analysis repetitions	200
Minimum absolute rotated loading	0.30
Minimum primitives / final factor	2
Parent construction	equal signed primitives

Monthly cross-sectional Spearman correlations are overlap-weighted through time. Exposure distance is $1 - |\rho|$, so a signal and its negation are treated as the same measurement and their sign is stored separately. HDBSCAN is used only for redundancy reduction. Final factors come from the common 131-month primitive-return window (2012-02-01 to 2022-12-01); no downloaded benchmark or known-factor return enters either stage. Beta is deliberately left in primitive returns so beta/BAB can appear as a latent risk dimension; beta neutrality belongs to later portfolio construction, not unsupervised grouping.

2 Training-only Sharpe admission

Each coverage-valid raw signal is first constructed as a standalone sector-demeaned, beta-neutral, signal-weighted long-short FMP after 10 bp turnover cost. Its direction is chosen from the 2012–2022 training Sharpe. It is admitted only when at least 70.0% of 2000 circular stationary-bootstrap paths, using an expected 12-month block, have positive Sharpe. This makes the extension supervised; no 2023+ return enters the screen.

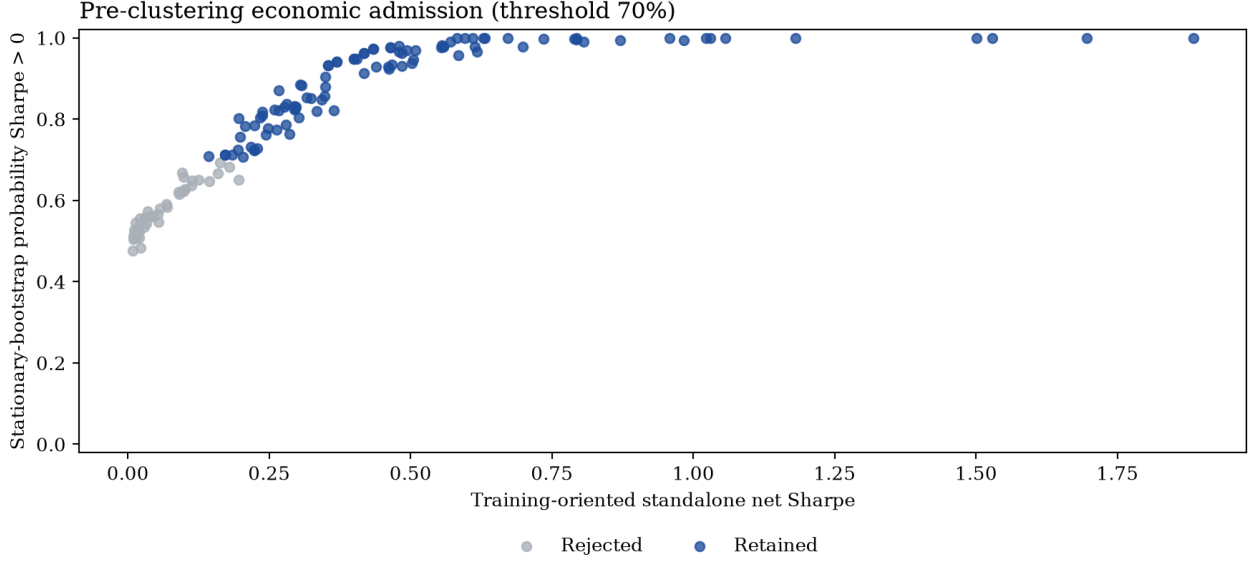


Figure 1: Standalone training Sharpe and block-bootstrap admission probability.

Rejected signal	N	Raw SR	Oriented SR	P(SR>0)	Reason
fcf_margin_ttm	131	0.16	0.16	69.3%	bootstrap_probability_below_threshold
is_negative_equity	131	-0.18	0.18	68.2%	bootstrap_probability_below_threshold
eps_growth_yoy_sign_flip	131	0.10	0.10	66.8%	bootstrap_probability_below_threshold
fcf_yield	131	0.16	0.16	66.6%	bootstrap_probability_below_threshold
info_discreteness_21_252	131	-0.10	0.10	65.8%	bootstrap_probability_below_threshold
roa	131	-0.13	0.13	65.1%	bootstrap_probability_below_threshold
equity_growth_yoy_sign_flip	131	0.20	0.20	65.1%	bootstrap_probability_below_threshold
book_to_market	131	0.11	0.11	64.8%	bootstrap_probability_below_threshold
cash_to_assets	131	-0.14	0.14	64.7%	bootstrap_probability_below_threshold
net_income_growth_yoy_ttm	131	0.11	0.11	63.7%	bootstrap_probability_below_threshold
operating_margin	131	0.10	0.10	62.7%	bootstrap_probability_below_threshold
buffett_ronta_proxy	131	0.10	0.10	62.3%	bootstrap_probability_below_threshold
beta_stability_252d_252v_1260c_SRY1	131	-0.10	0.10	62.3%	bootstrap_probability_below_threshold
cfoa	131	0.09	0.09	62.1%	bootstrap_probability_below_threshold
revenue_growth_yoy_sign_flip	131	-0.09	0.09	61.6%	bootstrap_probability_below_threshold
roc	131	-0.07	0.07	59.2%	bootstrap_probability_below_threshold
momentum_126_252	131	0.07	0.07	58.4%	bootstrap_probability_below_threshold
ivol_21d	131	-0.06	0.06	58.0%	bootstrap_probability_below_threshold
growth_roa_5y	131	-0.03	0.03	57.4%	bootstrap_probability_below_threshold
growth_roe_5y	131	0.05	0.05	56.6%	bootstrap_probability_below_threshold
rota_ebit	131	-0.04	0.04	56.1%	bootstrap_probability_below_threshold
gross_profit_growth_yoy_ttm	131	0.04	0.04	56.0%	bootstrap_probability_below_threshold
roe	131	-0.04	0.04	56.0%	bootstrap_probability_below_threshold
is_negative_net_operating_assets	131	-0.03	0.03	55.7%	bootstrap_probability_below_threshold
gross_margin	131	-0.02	0.02	55.6%	bootstrap_probability_below_threshold
vol_63d	131	0.05	0.05	54.8%	bootstrap_probability_below_threshold
is_positive_income_tax	131	-0.01	0.01	54.5%	bootstrap_probability_below_threshold
rd_to_revenue	131	-0.03	0.03	54.2%	bootstrap_probability_below_threshold

Rejected signal	N	Raw SR	Oriented SR	P(SR>0)	Reason
is_negative_working_capital	131	-0.03	0.03	53.4%	bootstrap_probability_below_threshold
operating_profitability	131	-0.02	0.02	53.4%	bootstrap_probability_below_threshold
high_52w_252d	131	-0.01	0.01	52.6%	bootstrap_probability_below_threshold
interest_to_opinc	131	0.01	0.01	52.6%	bootstrap_probability_below_threshold
earnings_to_price	131	-0.02	0.02	52.2%	bootstrap_probability_below_threshold
tax_rate	131	-0.01	0.01	51.4%	bootstrap_probability_below_threshold
growth_cfoa_5y	131	0.02	0.02	51.0%	bootstrap_probability_below_threshold
fcf	131	-0.02	0.02	50.8%	bootstrap_probability_below_threshold
is_positive_net_income	131	-0.01	0.01	50.8%	bootstrap_probability_below_threshold
net_margin	131	-0.01	0.01	50.4%	bootstrap_probability_below_threshold
is_negative_ev	131	0.02	0.02	48.2%	bootstrap_probability_below_threshold
momentum_21_63	131	0.01	0.01	47.7%	bootstrap_probability_below_threshold

3 Exposure-redundancy primitives

HDBSCAN produced 62 primitives: 26 multi-signal groups and 36 singletons. This is not treated as the final factor count; the earlier HDBSCAN-only interpretation failed because a dense redundancy map naturally preserves many unrelated signals as noise.

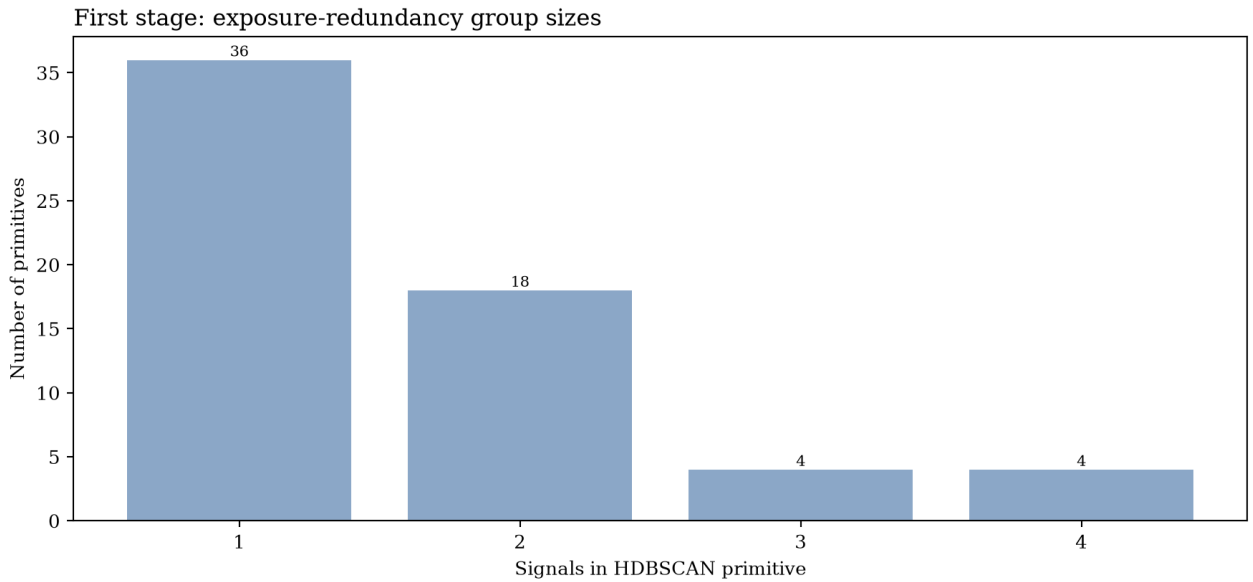


Figure 2: Distribution of signal counts in the exposure-redundancy primitives.

4 Latent factor count

Parallel analysis compares observed correlation-matrix eigenvalues with the 95th percentile from independently circular-shifted primitive returns. It retains 5 dimensions; 5 remain after the loading and minimum-signal rules.

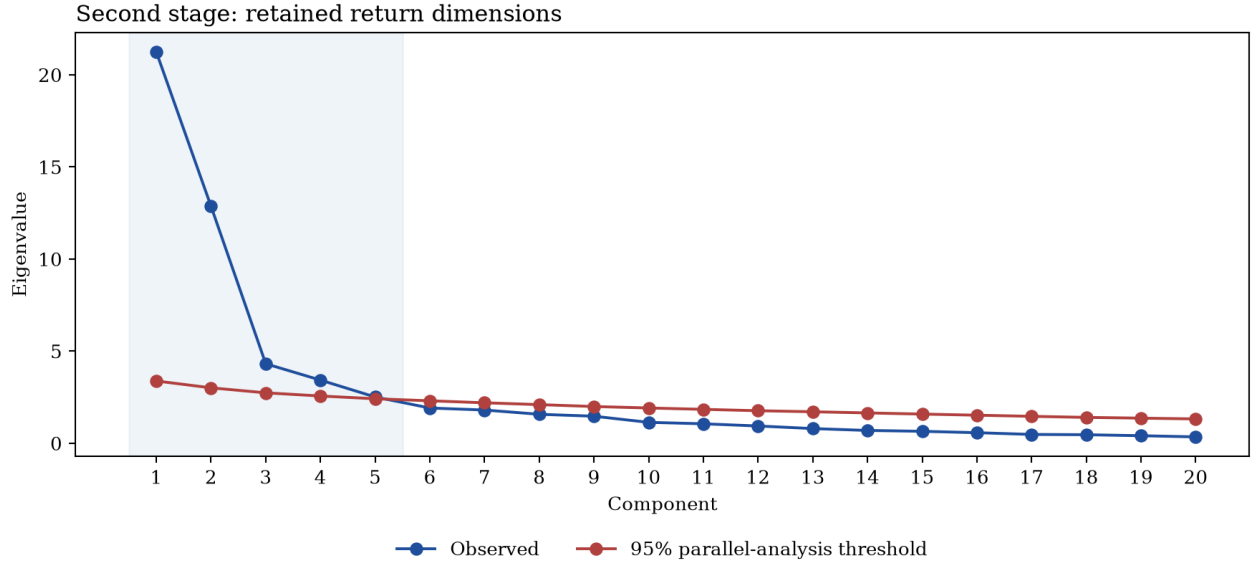


Figure 3: Observed and simulated eigenvalues. The shaded components are retained.

5 Rotated structure

Varimax rotation makes each primitive load mainly on one component. Loading sign orients a primitive within its factor; loading magnitude is not used as a portfolio weight. Each parent gives every signed primitive composite one equal vote, regardless of how many raw signals that primitive contains.

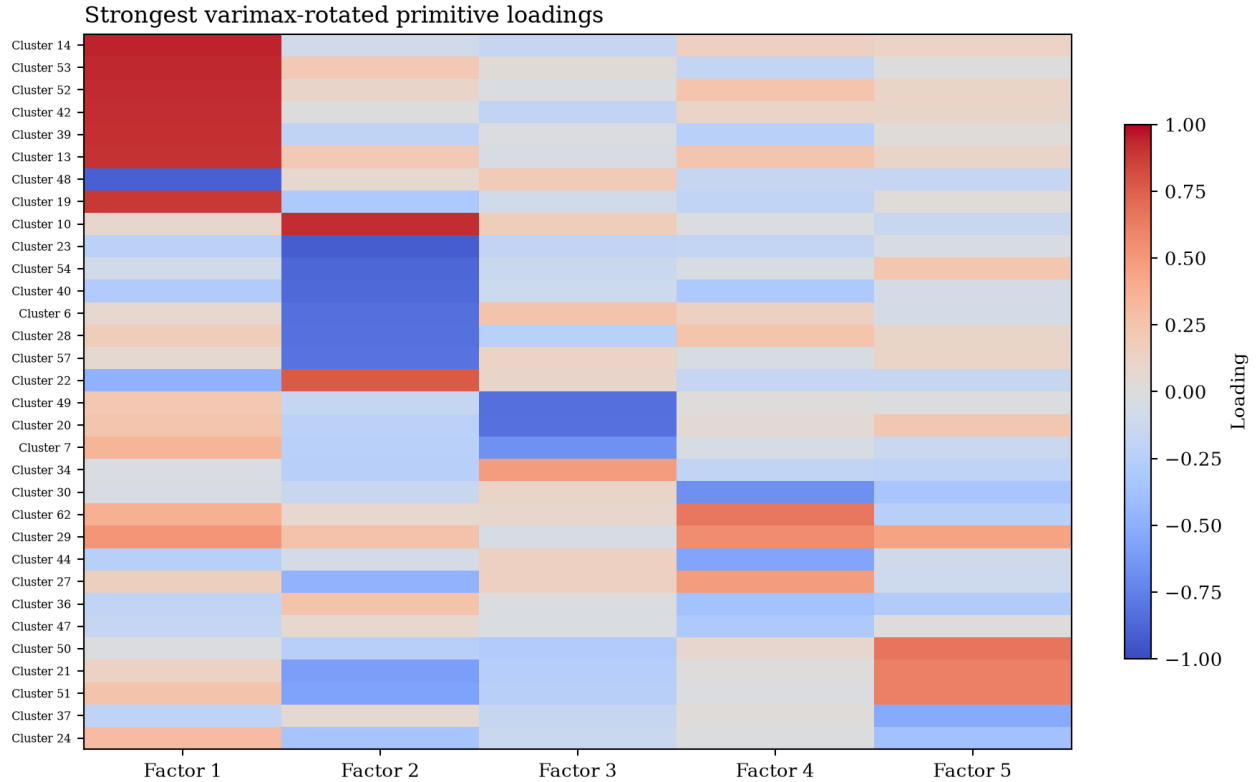


Figure 4: Rotated loading matrix, showing up to eight strongest retained primitives per factor.

6 Final candidate factors

The signs below give orientation only. When the Sharpe screen is enabled, economic evidence admits raw signals before clustering; no later rule removes a grouped signal or factor. The signed-signal table is the final equal-primitive construction.

Factor	Primitives	Raw signals	Signed primitive groups	Final signed signals
Factor 1	27	50	+Cluster 14, +Cluster 53, +Cluster 52, +Cluster 42, +Cluster 39, +Cluster 13, -Cluster 48, +Cluster 19, +Cluster 15, -Cluster 58, +Cluster 33, -Cluster 60, -Cluster 38, +Cluster 12, +Cluster 32, -Cluster 61, +Cluster 9, -Cluster 45, +Cluster 5, +Cluster 17, -Cluster 25, +Cluster 41, +Cluster 1, +Cluster 3, +Cluster 4, +Cluster 2, +Cluster 26	+bab 252v 1260c FRENCH CRSP, +bab shrunk 252v 1260c FRENCH CRSP, -beta 252v 1260c FRENCH CRSP, -beta shrunk 252v 1260c FRENCH CRSP, +bab 252v 1260c SPY, +bab shrunk 252v 1260c SPY, -beta 252v 1260c SPY, -beta shrunk 252v 1260c SPY, +bab 252v 1260c VTI, +bab shrunk 252v 1260c VTI, -beta 252v 1260c VTI, -beta shrunk 252v 1260c VTI, +ebit growth yoy ttm, +eps growth yoy, +net income growth yoy, +net income growth yoy sign flip, +accruals, +accruals percent, +accru- als total, -amihud illiq 21d, +dollar volume 21d, +corr 1260d FRENCH CRSP, +corr 1260d VTI, +earnings yield, +ebit to ev, +ebit margin ttm, +net income margin ttm, +fcf to eq- uity, +fcf to market cap, +gp to eq- uity, +gpoa, +log market cap, +mar- ket cap, -ret volume corr 63d, -skew 63d, +surprise, +surprisePercentage, +cash earnings ttm, +corr 1260d SPY, -equity issuance yoy, +ev, +fcf growth yoy ttm, +fcf margin, -interest to debt, -lottery max 5 21d, +net income to market cap, +net operating assets proxy, -roe volatility, -sga to revenue, -share turnover 21d
Factor 2	18	28	+Cluster 10, -Cluster 23, - Cluster 54, -Cluster 40, - Cluster 6, -Cluster 28, - Cluster 57, +Cluster 22, - Cluster 18, -Cluster 55, +Clus- ter 16, +Cluster 8, +Cluster 46, -Cluster 11, -Cluster 59, -Cluster 43, +Cluster 31, - Cluster 56	-accruals noa proxy, -asset growth yoy, -equity growth yoy, +debt to assets, +debt to equity, -ncav to mc, +as- sets to mc, -ev to assets, -cash ra- tio, -current ratio, +gp ev, +gross profit to ev, -growth gmar 5y, -growth gpoa 5y, -momentum 21 756, +value proxy 1260d, -ps ttm, +sales to price, -altman z, +capital intensity, -evs ttm, -gross profit margin ttm, +is negative invested capital, -pb, -pe ttm, -rd to opex, -revenue growth yoy, -roic nopat
Factor 3	4	7	-Cluster 49, -Cluster 20, - Cluster 7, +Cluster 34	-bb pctb 20d, -rsi 14d, -stoch k 14d, - ma dist 63d, -price zscore 63d, +debt issuance yoy, -momentum 0 21
Factor 4	7	7	-Cluster 30, +Cluster 62, +Cluster 29, -Cluster 44, +Cluster 27, -Cluster 36, - Cluster 47	+accruals wc proxy, +asset turnover, - capex to revenue, -dpo, -growth acc 5y, -is positive revenue, +working capital proxy
Factor 5	5	7	+Cluster 50, +Cluster 21, +Cluster 51, -Cluster 37, - Cluster 24	+momentum 21 252, +resid momen- tum 21 252, -resid ret 252v 1260c SPY, -ret zscore 63d, -dso, +momentum 21 126, +momentum 21 252 vol 63d

7 Final factor FMP performance

Each factor is constructed once from its frozen component exposures using the configured rank transformation, composite-level sector demeaning, signal-weighted long-short portfolio and factor-level beta neutrality. The curves

and statistics use the same complete common return window.

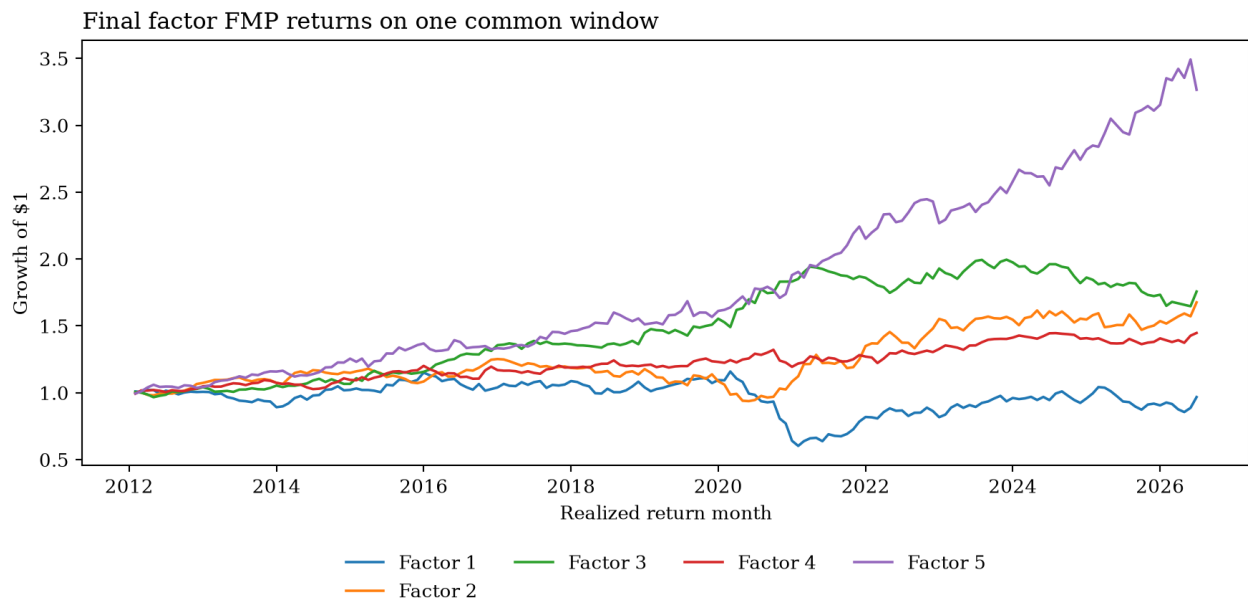


Figure 5: Net cumulative returns of the final factor books.

Factor	N	Start	End	Ann. return	Ann. vol.	Sharpe	HAC t	Max DD
Factor 1	174	2012-02-01	2026-07-01	0.48%	11.75%	0.04	0.14	-48.02%
Factor 2	174	2012-02-01	2026-07-01	3.95%	8.78%	0.45	1.57	-25.12%
Factor 3	174	2012-02-01	2026-07-01	4.12%	6.81%	0.60	2.42	-17.47%
Factor 4	174	2012-02-01	2026-07-01	2.69%	5.23%	0.51	2.34	-9.66%
Factor 5	174	2012-02-01	2026-07-01	8.52%	8.19%	1.04	5.43	-7.33%

8 Interpretation and reproducibility

This remains an exploratory return-risk taxonomy: the performance table describes the resulting books but does not select or eliminate factors. Exposure discovery run 20260906T011105Z_e8fdca00a5. The run directory preserves primitive definitions, raw-signal lineage, rotated loadings, stock-level final exposures, primitive and final FMP returns, configuration and manifest for Factor Runner or HMM reuse.